

Total No. of Questions: 07

B.Sc.(IT) (2015 & Onward) / BCA (2011 & Onwards) (Sem.– 2)

MATHEMATICS – II

M Code: 10051

Subject Code: BSIT/BSBC-202

Paper ID: [B1114]

Time: 3 Hrs.

Max. Marks: 60

INSTRUCTIONS TO CANDIDATES:

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains SIX questions carrying TEN marks each and students has to attempt any FOUR questions.

SECTION A

1. a) If $A = \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}$ and $B = (-2 \ -1 \ -4)$, then verify that $(AB)^T = B^T A^T$, where T stands for transpose.
- b) Find for the value of x for which the matrix $\begin{pmatrix} 1 & -2 & 3 \\ 1 & 2 & 1 \\ x & 2 & -3 \end{pmatrix}$ is a singular matrix.
- c) Find the rank of the matrix $\begin{pmatrix} 2 & 3 & 4 & -1 \\ 5 & 2 & 0 & -1 \\ -4 & 5 & 12 & -1 \end{pmatrix}$
- d) The mean of 200 items was 50. Later on it was found that two items were misread as 92 and 8 instead of 192 and 88. Find out the correct mean.
- e) Differentiate $\sin x^\circ$ with respect to x .
- f) Evaluate $\int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$.
- g) Obtain the value of the integral $\int_0^{\pi/2} \frac{1}{1 + \cos \theta} d\theta$.
- h) Calculate mean deviation about mean from the following data:

$x:$	3	9	17	23	27
$f:$	8	10	12	9	5
- i) State Simpson's $1/3^{\text{rd}}$ and $3/8^{\text{th}}$ rule to evaluate $\int_a^b f(x) dx$.
- j) If $\log_5 (x^2 + x) + \log_5 (x + 1) = 2$, then find the value of x .

SECTION B

2. a) Use Cramer's rule to solve the system of equations:

$$x - 4y - z = 11; 2x - 5y + 2z = 39; -3x + 2y + z = 1 \quad (5)$$

b) Find the inverse of the matrix $A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}$ (5)

3. a) If $A = \begin{pmatrix} 3 & 2 & 0 \\ 1 & 4 & 0 \\ 0 & 0 & 5 \end{pmatrix}$ then show that $A^2 - 7A + 10I_3 = 0$ (5)

- b) How much will Rs. 20,000 amount to, in 3 years at compound interest if the rates for successive years are 5%, 8% and 10% per year. (5)

4. a) If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, then show that $\frac{dy}{dx} = -\frac{1}{(1+x)^2}$ (5)

- b) Show that of all the rectangles of given area, the square has the smallest perimeter.

5. a) If $x = \frac{1-t^2}{1+t^2}$ and $y = \frac{2t}{1+t^2}$ then find the value of $\frac{dy}{dx}$ at $t = 1$. (5)

b) Evaluate the integral $\int \frac{x^3}{(x-1)(x-2)} dx$ (5)

6. a) Evaluate $\int_0^5 \frac{1}{4x+5} dx$ by using Simpson's 1/3 rule by dividing the range into 10 equal parts, and compare the result. (5)

- b) Differentiate the following w.r.t. x

i) $\frac{e^x + e^{-x}}{e^x - e^{-x}}$ ii) $\log(x + \sqrt{a^2 + x^2})$ (5)

7. a) Goals scored by two teams A and B in a football season were as under:

Number of Goals scored	Number of matches	
In a match	A	B
0	27	17
1	9	9
2	8	6
3	5	5
4	4	3

Test, which team is more consistent in scoring goals. (6)

- b) Find the value of median of the following frequency distribution:

x :	1	2	3	4	5	6	7	8	9	(4)
f :	8	10	11	16	20	25	15	9	6	

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BCA (Sem. – 2)
MATHEMATICS-II
Subject Code: BSBC/BSIT-202
M.Code : 10051
Date of Examination : 15-12-22

Time: 3 Hrs.

Max. Marks: 60

INSTRUCTIONS TO CANDIDATES:

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains SIX questions carrying TEN marks each and students have to attempt any FOUR questions.

SECTION A

1. Answer the following:

a) If $\begin{bmatrix} 2x+1 & 2y \\ 0 & y^2+1 \end{bmatrix} = \begin{bmatrix} x+3 & 10 \\ 0 & 26 \end{bmatrix}$

Write the value of $(x + y)$.

b) Find the values of x for which $\begin{vmatrix} 3 & x \\ x & 1 \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix}$

c) Differentiate $e^{\log(x+\sqrt{x^2+u^2})}$ w.r.t x .

d) If $\sin x = y \sin(x + b)$. Show that $\frac{dy}{dx} = \frac{\sin b}{\sin^2(x+b)}$.

e) Define Median and Standard deviation with example.

f) Calculate the mean deviation from median and coefficient of mean deviation

20, 22, 25, 38, 40, 50, 65, 70, 75

g) Evaluate $\int \frac{x^3-1}{x^2} dx$.

h) Find $\int \sin^3 x dx$.

i) Evaluate $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^3 x dx$.

j) State Simpson's $\frac{1}{3}$ rd rule.

SECTION B

2. For the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$ Show that $A^3 - 6A^2 + 5A + 11I = 0$

Hence find A^{-1}

3. Using matrices solve the following system of equations

$$3x + 4y + 7z = 4$$

$$2x - y + 3z = -3$$

$$x + 2y - 3z = 8$$

4. a) Determine two positive numbers whose sum is 15 and the sum whose squares is minimum.

b) If $y = (x + \sqrt{x^2 + a^2})^n$ Prove that $\frac{dy}{dx} = \frac{ny}{\sqrt{x^2 + a^2}}$.

5. a) Find $\int x \sin^{-1} x dx$

b) Evaluate $\int (3 - 2x)\sqrt{2 + x - x^2} dx$.

6. a) Evaluate $\int_0^6 \frac{1}{1+x^2} dx$ by using Simpson's $\frac{1}{3}^{rd}$ rule.

- b) Calculate the Median of the following data

Marks:	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100
No. of Students :	0	5	22	25	16

7. a) Find the Standard deviation and Mean for the distribution of wages of 65 employees working in a factory

Wages (Rs) :	55	65	75	85	95	105	115
No. of employees :	8	10	6	14	10	5	2

- b) Write the formula for calculating Mode, Median in continuous series.

SECTION-B

2. Solve the following equation by Matrix – Inversion method

$$5x + 3y + z = 16$$

$$2x + y + 3z = 19$$

$$x + 2y + 4z = 25$$

3. Calculate Mean and Mode of following data

Class	6 – 7	7 – 4	14 – 21	21 – 28	28 – 35	35 – 42
Frequency	19	25	36	72	51	45

4. a) Differentiate the following wrt x

i) $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ ii) $\frac{e^x + \sin 2x}{1 + \cos 2x}$

b) Evaluate $\int \frac{dx}{(x+1)(x+2)}$.

5. Evaluate $\int_0^4 e^x dx$ by Simpson's $\frac{1}{3}$ rd rule and Simpson's $\frac{3}{8}$ th rule. Given that $e = 2.72$, $e^2 = 7.39$, $e^3 = 20.09$, $e^4 = 24.6$.
6. Show that of all the rectangles of given areas the square has the smallest perimeter.
7. Evaluate $\int e^x(1+x) \log(xe^x) dx$.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.